Plain LaTeX Through texwright

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October 7, 2026

Abstract

This article is ordinary LaTeX: the same source compiles with `pdflatex`. texwright renders it without a TeX installation, numbering everything the way LaTeX would and reporting anything it cannot handle.

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1 Definitions

Definition 1. The trace of a square matrix $A \in \mathbb{R}^{n\times n}$ is $\operatorname{tr} A = \sum_i a_{ii}$.

Lemma 1.1. *For conformable matrices,* $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. Both sides equal $\sum_{i,j} a_{ij} b_{ji}$. □

Theorem 1.2 (Similarity invariance). *If* $B = P^{-1}AP$ *then* $\operatorname{tr} B = \operatorname{tr} A$.

Proof. By Lemma 1.1, $\operatorname{tr}(P^{-1}(AP)) = \operatorname{tr}((AP)P^{-1}) = \operatorname{tr} A$. □

2 Equations

The heat equation and its boundary conditions are

$$u_t = \alpha u_{xx}, \tag{1}$$

$$u(0,t) = u(1,t) = 0. \tag{2}$$

Equation (1) is due to Fourier [1]; the standard reference is [2, ch. 2]. Absolute values: $|x - y| \le |x| + |y|$.

3 Results

Table 1 reports errors of the scheme of Crank and Nicolson [3]; the profiles are in Figure 1, and Theorem 1.2 is in Section 1 on page 1.

- Second order in time and space.[1]
- Unconditionally stable.

[1] For smooth initial data.

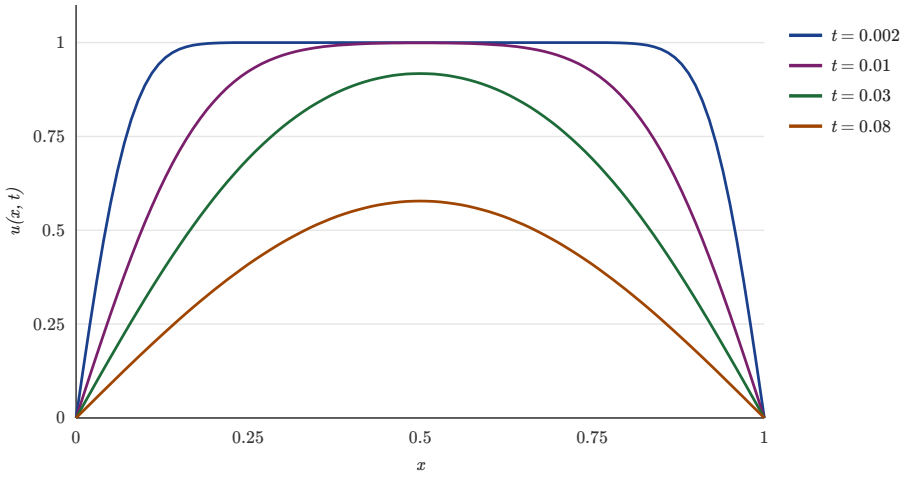


Figure 1: Solutions of (1) at four times.

Table 1: Crank–Nicolson errors at $t = 0.1$.

Grid		**Error**	
N	Δt	max	rate
16	10^{-2}	3.1×10^{-3}	—
32	5×10^{-3}	7.8×10^{-4}	1.99
64	2.5×10^{-3}	1.9×10^{-4}	2.00

References

[1] J.-B. J. Fourier, *Théorie analytique de la chaleur*. Paris: Firmin Didot, 1822.

[2] L. C. Evans, *Partial Differential Equations*, 2nd ed. Providence, RI: American Mathematical Society, 2010.

[3] J. Crank and P. Nicolson, “A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type,” *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 43, no. 1, pp. 50–67, 1947, doi: 10.1017/S0305004100023197.